

THE PACODU VAULT

An AI-Assisted Conceptual Framework for Cost-Escalation Authentication and Adaptive Defensive Architecture

Author: C. Pancu

Academic Background: BSc Information Technology, Communications & Networking

Project Acronym: PaCoDu

Document Classification: AI-Assisted Conceptual Research White Paper

License: Creative Commons Attribution-NonCommercial 4.0 International (CC BY-NC 4.0)

AUTHOR'S FOREWORD

The PaCoDu Vault did not begin with the intention of inventing a new cryptographic primitive or proposing an unbreakable authentication protocol. It began with a much simpler question:

If every authentication system can ultimately be compromised given sufficient time, resources, expertise, and determination, is the pursuit of "unhackable" security the most useful objective?

As someone with a BSc in Information Technology, Communications and Networking, I possess academic exposure to networking, infrastructure, operating systems, and information security. However, I am not formally trained as a cryptographer, hardware security engineer, theoretical computer scientist, or distributed systems researcher.

The central idea presented in this paper emerged not from deep expertise in any single discipline, but from systems-level reasoning. The original concept was progressively explored through extensive collaboration with advanced artificial intelligence systems, which assisted in

formalizing terminology, organizing architectural concepts, improving technical language, identifying weaknesses, exploring mathematical representations, and examining possible implementation strategies.

The resulting document therefore represents neither purely human authorship nor autonomous AI generation. Rather, it documents an iterative collaborative design process in which human intuition defined the research direction while AI served as a technical reasoning and editorial partner.

This distinction is important.

The contribution claimed by this paper is not the invention of new cryptographic algorithms, authentication primitives, or security mathematics. Instead, the contribution lies in proposing an alternative architectural philosophy for authentication systems—one that evaluates defensive effectiveness not solely by resistance to compromise, but by the cumulative operational cost imposed upon those attempting that compromise.

Accordingly, this document should not be interpreted as a finalized protocol specification, production-ready authentication system, or peer-reviewed scientific result. It is presented as an AI-assisted conceptual architectural framework intended to stimulate discussion, encourage interdisciplinary collaboration, invite critical review, and provide a foundation upon which future academic and engineering research may build.

ABSTRACT

Modern authentication systems have achieved remarkable advances through stronger cryptographic algorithms, hardware-rooted trust, adaptive authentication, and increasingly sophisticated defensive mechanisms. Nevertheless, many security architectures continue to rely upon authentication artifacts and infrastructure components that remain observable, reusable, or economically attractive targets over extended operational periods.

The PaCoDu Vault explores an alternative design philosophy.

Rather than pursuing the unattainable objective of creating an authentication system that can never be compromised, the framework proposes increasing the cumulative resources required to compromise it. The underlying hypothesis is that authentication architectures should be evaluated not only by whether compromise is theoretically possible, but also by the operational, computational, financial, temporal, and organizational costs imposed upon an adversary attempting that compromise.

To investigate this hypothesis, the framework combines established concepts including multimodal biometric authentication, fuzzy extractors, hardware-protected trust anchors, HMAC-derived temporal authentication, moving-target infrastructure, deception-oriented defensive mechanisms, and adaptive orchestration into a unified conceptual architecture. Each architectural layer is intended to increase one or more dimensions of attacker cost while reducing the long-term value of persistent reconnaissance and credential compromise.

The framework intentionally incorporates both established engineering practices and exploratory research directions. It does not claim cryptographic novelty, mathematical proof of security, production readiness, or superiority over existing authentication systems. Instead, it should be understood as an AI-assisted conceptual architectural proposal that seeks to encourage discussion regarding a complementary philosophy of authentication design based upon economic deterrence through cumulative cost escalation.

Ultimately, the PaCoDu Vault asks a different question from many traditional security architectures. Instead of asking, *"Can an authentication system become impossible to compromise?"*, it asks, *"Can authentication systems be designed so that compromise becomes progressively less worthwhile?"*

If the answer is affirmative, then the effectiveness of future defensive architectures may be measured not only by the strength of their cryptography, but also by their ability to reshape attacker behaviour through sustained increases in operational cost.

1. INTRODUCTION

Over the past several decades, authentication technologies have evolved considerably through advances in cryptography, hardware security, biometrics, adaptive authentication, and distributed defensive architectures. Modern systems routinely combine multiple layers of protection, making unauthorized access significantly more difficult than in previous generations of identity management.

Despite these advances, a common characteristic remains present across many authentication mechanisms: they rely upon verification artifacts, infrastructure components, or operational states that persist long enough to become attractive targets for observation, reconnaissance, interception, replay, or systematic analysis.

Absolute security remains an aspirational objective rather than a practical guarantee. Given sufficient resources, expertise, time, and opportunity, any defensive system may eventually be compromised through technical exploitation, operational mistakes, insider threats, supply-chain attacks, or physical access.

The PaCoDu Vault does not attempt to solve this fundamental limitation.

Instead, it proposes that authentication systems should be evaluated from an additional perspective: the economics of attack.

Rather than asking whether compromise is theoretically possible, this framework asks whether defensive architectures can be intentionally designed so that successful compromise requires progressively greater investments of time, expertise, financial resources, computational capability, coordination, and operational risk.

Under this perspective, security is viewed not solely as resistance to attack, but also as the systematic escalation of attacker cost.

The PaCoDu framework explores one possible architectural approach toward achieving that objective.

2. RESEARCH MOTIVATION

The motivation behind this work originated from a simple observation rather than from advances in cryptography.

Throughout the history of cybersecurity, defensive technologies have continuously improved. At the same time, attackers have adapted their methods, tools, and organizational structures in response. This ongoing cycle suggests that cybersecurity is not a problem that can be permanently solved through increasingly sophisticated defensive mechanisms alone.

Instead, it represents a continuous strategic competition between defenders and adversaries.

Within this competition, attackers generally seek to maximise return while minimising expenditure. A successful attack is typically characterised by low operational cost, limited time investment, modest equipment requirements, minimal collaboration, and a high probability of reward.

Conversely, defenders often focus on strengthening cryptographic algorithms, improving authentication accuracy, increasing computational complexity, or adding additional protective layers. While these measures remain essential, they do not always explicitly consider the broader economic incentives influencing attacker behaviour.

This observation motivated the central question explored throughout this paper:

Can authentication architectures be deliberately designed to alter the economics of attack by increasing the cumulative resources required for successful compromise?

Rather than attempting to eliminate attacks entirely, such architectures may instead reduce the attractiveness of attacking them by increasing the operational burden imposed upon adversaries.

The PaCoDu Vault represents one conceptual exploration of this alternative design philosophy.

3. CORE HYPOTHESIS

The central hypothesis of the PaCoDu framework is that the effectiveness of an authentication system should not be evaluated solely according to whether compromise is theoretically possible.

Instead, defensive architectures should also be evaluated according to the cumulative cost they impose upon an adversary attempting that compromise.

Within this framework, attacker cost extends beyond computational complexity alone and includes multiple independent dimensions:

- Financial investment required to perform the attack.
- Time required for reconnaissance, preparation, execution, and persistence.
- Specialist knowledge across multiple technical disciplines.
- Hardware capabilities necessary for successful exploitation.
- Computational resources required to sustain the attack.
- Organisational coordination between multiple participants.
- Operational exposure to detection, attribution, and defensive response.

The hypothesis therefore proposes that authentication systems become increasingly resilient—not because they are impossible to compromise—but because successful compromise progressively exceeds the resources that many adversaries are willing or able to invest.

Under this interpretation, resilience emerges from cumulative cost escalation rather than from the assumption of absolute invulnerability.

4. DESIGN PHILOSOPHY

The PaCoDu framework is built upon a single guiding principle:

Perfect security should not be considered the primary objective of authentication architecture. The primary objective should be to maximise the cumulative cost of successful compromise while preserving practical usability for legitimate users.

This philosophy does not replace conventional cybersecurity principles.

Cryptographic strength, secure hardware, behavioural authentication, infrastructure redundancy, and continuous monitoring remain essential components of modern defensive systems.

Instead, PaCoDu proposes complementing these established practices with an additional design objective: deliberately engineering each defensive layer so that it increases one or more dimensions of attacker cost.

Within this framework, every architectural component serves a dual purpose.

First, it contributes to the direct protection of authentication assets.

Second, it contributes to increasing the resources required to defeat subsequent defensive layers.

Consequently, the framework should be understood not as a collection of independent security mechanisms, but as a coordinated cost-escalation architecture in which successive defensive layers progressively increase uncertainty, complexity, operational effort, financial investment, specialised expertise, and time requirements for potential adversaries.

The objective is not to create an authentication system that can never be compromised.

The objective is to create an authentication system for which compromise becomes progressively less attractive, less economical, and less practical as additional defensive layers accumulate.

This distinction represents the fundamental philosophical contribution proposed throughout the remainder of this document.

5. ARCHITECTURAL MODEL

The PaCoDu Vault is organized as a layered authentication architecture composed of five complementary defensive rings. Each ring addresses a different aspect of authentication security while simultaneously contributing to the central objective of the framework: increasing the cumulative cost of successful compromise.

Unlike traditional defence-in-depth models, where each layer primarily attempts to prevent intrusion, the PaCoDu architecture assigns a second responsibility to every defensive component. Beyond providing technical protection, each layer is intentionally designed to increase one or more dimensions of attacker cost, including time, financial investment, computational resources, specialist expertise, organisational coordination, and operational risk.

No individual ring is assumed to be impossible to defeat. Instead, the framework proposes that resilience emerges from the cumulative interaction of multiple independent defensive mechanisms. As an adversary progresses through the architecture, each successive ring introduces new technical challenges and additional resource requirements, making successful compromise progressively more demanding.

The five rings should therefore be understood not as isolated security mechanisms, but as complementary components of a single cost-escalation strategy.

Ring 1 — Defensive Perimeter Layer

The first ring represents the externally visible defensive boundary of the authentication infrastructure. Distributed edge services perform request filtering, anomaly detection, rate limiting, traffic inspection, and distributed denial-of-service mitigation before authentication requests reach the internal verification pipeline.

Anycast distribution and geographically dispersed infrastructure improve availability while reducing dependence upon individual network locations.

Within the PaCoDu philosophy, however, the primary contribution of this layer extends beyond conventional perimeter defence. By increasing infrastructure dispersion and reducing predictability, the ring forces adversaries to invest additional time and computational resources simply to perform reconnaissance or establish consistent communication with the authentication services.

Rather than preventing every attack, this layer seeks to increase the operational cost associated with reaching the deeper authentication mechanisms.

Ring 2 — Behavioural and Cognitive Authentication Layer

The second ring supplements conventional authentication by incorporating behavioural and cognitive interaction alongside traditional identity verification.

Potential authentication factors include conditioned semantic associations, behavioural response sequences, recognition tasks, and other interaction patterns that are naturally difficult to reproduce without prolonged observation.

The objective is not to replace established authentication methods, but to reduce reliance upon static credentials alone. Behavioural characteristics introduce additional uncertainty because they often require attackers to observe legitimate users over extended periods while simultaneously reproducing subtle interaction patterns with sufficient consistency.

Consequently, this layer primarily increases the time, observation effort, and specialist expertise required to perform successful impersonation.

Ring 3 — Biometric Trust Layer

The third ring establishes the cryptographic trust foundation of the architecture through multimodal biometric authentication.

Rather than relying upon a single biometric modality, the framework explores combining multiple independent biological characteristics—including palm-vein topology, electrocardiographic signatures, acoustic characteristics, or similar biometric measurements—into a unified authentication vector.

Because biological measurements naturally contain variability, the architecture references fuzzy extractor principles to derive stable cryptographic material from imperfect biometric observations. The resulting key material is intended to remain isolated inside hardware-protected execution environments such as Trusted Execution Environments (TEEs), Secure Enclaves, or Hardware Security Modules (HSMs).

By requiring simultaneous compromise of biological authentication mechanisms and hardware-rooted trust anchors, this layer significantly increases both the technical sophistication and equipment requirements associated with credential theft.

Ring 4 — Temporal Authentication Layer

The fourth ring introduces continuous temporal variation into the authentication process.

Stable cryptographic material derived from the previous layer is used to generate short-lived authentication assertions based upon time-dependent HMAC calculations. Unlike conventional authentication tokens that may remain valid for relatively long periods, PaCoDu proposes continuously regenerating authentication assertions at short time intervals.

This approach seeks to reduce the operational value of intercepted authentication data by ensuring that credentials rapidly become obsolete after generation.

Recognising the practical realities of distributed computing, the framework also incorporates a bounded sliding validation window that tolerates normal network latency, packet jitter, processing overhead, and minor clock drift. This optimisation preserves the security benefits of short-lived authentication while improving reliability under real-world operating conditions.

The result is a temporal authentication model that balances aggressive credential mutation with practical deployability across distributed infrastructure.

Ring 5 — Adaptive Infrastructure and Deception Layer

The fifth ring extends authentication beyond identity verification by incorporating adaptive infrastructure behaviour and deception-oriented defensive mechanisms.

The framework explores the use of honeytokens, decoy services, moving-target infrastructure, and adaptive orchestration strategies intended to reduce infrastructure predictability while simultaneously improving defensive visibility.

Rather than assuming that every intrusion attempt can be prevented, this layer accepts that adversaries may occasionally penetrate individual defensive mechanisms. Its objective is therefore to increase uncertainty by forcing attackers to distinguish genuine services from intentionally deceptive resources while continuously adapting to changing infrastructure layouts.

At the same time, interactions with deceptive infrastructure provide valuable telemetry that may improve detection capabilities and support defensive analysis during active intrusion attempts.

Within the broader PaCoDu philosophy, this ring primarily increases reconnaissance effort, analytical complexity, operational uncertainty, and the resources required to maintain persistent attacks against the authentication infrastructure.

Architectural Summary

Although each defensive ring performs a distinct technical function, the architecture should be understood as a unified defensive ecosystem rather than a collection of independent security mechanisms.

The effectiveness of the framework does not depend upon the absolute strength of any single layer. Instead, resilience emerges from the cumulative interaction of all five rings, each increasing different dimensions of attacker cost while complementing the capabilities of the others.

The PaCoDu Vault therefore proposes that authentication systems can become more resilient not because they are impossible to compromise, but because successful compromise progressively demands greater time, expertise, computational capability, financial investment, organisational coordination, and operational risk than many adversaries are willing or able to sustain.

6. AUTHENTICATION PIPELINE

The authentication pipeline represents the operational implementation of the architectural principles described in the previous chapter. Each stage contributes to the generation, validation, and continuous renewal of authentication assertions while simultaneously supporting the broader objective of increasing the cumulative cost of successful compromise.

The pipeline is intentionally modular. Individual components are based upon established security concepts wherever possible, while their combination reflects the conceptual contribution proposed by the PaCoDu framework.

Although presented sequentially, several stages may execute concurrently depending upon implementation requirements.

Stage 1 — Multimodal Authentication Vector Construction

The authentication process begins with the construction of a unified authentication vector derived from multiple independent identity characteristics.

Rather than depending upon a single authentication factor, the framework explores combining biological and behavioural information into a composite representation capable of supporting stronger identity verification while reducing dependence upon static credentials.

Potential biometric inputs include palm-vein topology, electrocardiographic (ECG) characteristics, acoustic signatures, facial geometry, or other suitable biometric measurements. These may be complemented by behavioural interaction patterns, conditioned semantic associations, cognitive recognition tasks, or additional behavioural authentication factors.

Conceptually, the authentication vector may be represented as:

INPUT_VECTOR = Biometrics + Behavioural_Data

Because biological measurements naturally contain variability, exact matching is neither expected nor desirable. Instead, the framework references fuzzy extractor principles that allow stable cryptographic material to be reconstructed from sufficiently similar biometric observations.

Conceptually:

Gen(INPUT_VECTOR) → (SECRET_KEY, HELPER_DATA)

Rep(INPUT_VECTOR_NEW, HELPER_DATA) → SECRET_KEY

Provided that:

$\text{dist}(\text{INPUT_VECTOR}, \text{INPUT_VECTOR_NEW}) \leq \delta$

Within the PaCoDu philosophy, this stage contributes to cost escalation by increasing the complexity of identity replication. Rather than stealing a static credential, an adversary must simultaneously reproduce multiple independent biometric and behavioural characteristics with sufficient accuracy to regenerate identical cryptographic material.

Stage 2 — Temporal Token Generation

Once stable key material has been reconstructed inside a hardware-protected execution environment, it is used to generate short-lived authentication assertions.

Rather than maintaining long-lived authentication credentials, the framework proposes deriving authentication tokens from continuously advancing temporal states.

Conceptually:

TIME_STEP = floor(Current_Unix_Time / Window_Size)

TOKEN = HMAC_SHA_512(SECRET_KEY, TIME_STEP)

This process generates authentication assertions whose validity continuously changes over time. As the temporal window advances, previously generated tokens rapidly lose operational value.

Unlike conventional static credentials, intercepted authentication assertions cannot be reused indefinitely because their validity is inherently tied to a narrowly defined temporal interval.

Within the PaCoDu philosophy, temporal authentication contributes to attacker cost escalation by reducing the practical value of captured credentials while simultaneously increasing the precision required for successful replay attacks.

Stage 2.1 — Operational Time-Window Optimisation

An aggressive temporal authentication interval provides significant theoretical resistance against credential replay but introduces practical engineering challenges within distributed computing environments.

Normal internet communication is affected by propagation delay, routing variation, packet jitter, processing latency, and endpoint clock drift. Under a strict single-window validation model, these unavoidable characteristics may cause legitimate authentication attempts to fail despite the absence of malicious activity.

To improve operational robustness, the PaCoDu framework proposes a bounded sliding validation window.

Authentication tokens continue to be generated using a short temporal interval—for example, a conceptual 1.5-second generation cycle—but verification nodes are permitted to

validate authentication assertions generated during both the current temporal epoch (T) and the immediately preceding epoch (T-1).

This limited lookback mechanism absorbs normal network variability without substantially increasing the useful lifetime of captured authentication assertions.

The optimisation therefore balances two competing engineering objectives:

- Maintaining short-lived authentication assertions that minimise replay opportunities.
- Preserving reliable authentication under realistic distributed network conditions.

The precise duration of the temporal window should not be interpreted as a fixed implementation requirement. Instead, it represents a tunable engineering parameter whose optimal value depends upon deployment characteristics, expected network latency, endpoint capabilities, and acceptable security trade-offs.

Within the PaCoDu philosophy, this optimisation demonstrates that increasing attacker cost should not come at the expense of unacceptable usability for legitimate users.

Stage 3 — Adaptive Infrastructure Interaction

Following successful authentication, verified requests proceed into the adaptive infrastructure environment described by the five-ring architecture.

Rather than exposing a static service topology, future implementations may incorporate infrastructure mutation, adaptive routing strategies, deception-oriented services, and dynamic workload distribution.

The objective is not to create an unpredictable system for legitimate users, but to reduce long-term infrastructure predictability for potential adversaries performing reconnaissance or attempting persistent attacks.

The specific mechanisms employed within this stage remain implementation-dependent and should therefore be regarded as conceptual architectural directions rather than mandatory protocol requirements.

Pipeline Summary

The authentication pipeline combines established security concepts—including multimodal authentication, fuzzy extractors, hardware-protected key isolation, HMAC-based

temporal authentication, and adaptive infrastructure behaviour—into a unified conceptual workflow.

None of these individual mechanisms are presented as novel cryptographic inventions.

Instead, the conceptual contribution of the PaCoDu framework lies in their coordinated application toward a common architectural objective: increasing the cumulative operational cost of successful compromise while maintaining practical usability for legitimate users.

The pipeline therefore represents the operational expression of the cost-escalation philosophy introduced throughout this paper.

7. THREAT MODEL AND SECURITY ANALYSIS

The PaCoDu framework is not designed around the assumption that attacks can be completely eliminated. Instead, it assumes that sufficiently capable adversaries will continue to adapt their techniques in response to new defensive mechanisms.

Consequently, the objective of the framework is not absolute prevention, but the progressive escalation of attacker cost throughout the authentication lifecycle.

This chapter identifies representative threat classes that the proposed architecture seeks to influence and discusses the extent to which the framework may increase the resources required for successful compromise.

Hardware Interdiction

Authentication systems that rely upon hardware-protected trust anchors remain susceptible to physical compromise of endpoint devices.

Attackers capable of obtaining prolonged physical access may attempt hardware probing, side-channel analysis, fault injection, memory extraction, or other techniques intended to recover cryptographic material before it enters protected execution environments.

The PaCoDu framework does not claim immunity against these attacks.

Instead, it proposes increasing the complexity of successful compromise by combining hardware-protected key storage with multimodal authentication, thereby requiring adversaries to defeat multiple independent trust mechanisms rather than a single credential repository.

Within the proposed philosophy, this threat primarily increases the required investment in specialised equipment, laboratory capability, and technical expertise.

Temporal Replay and Synchronisation Attacks

Short-lived authentication assertions reduce the operational value of intercepted credentials but simultaneously introduce sensitivity to network latency, clock drift, and processing delay.

The bounded sliding validation window proposed in Stage 2.1 attempts to balance these competing engineering requirements by permitting limited temporal tolerance while maintaining relatively short credential lifetimes.

Although this optimisation cannot eliminate replay attempts entirely, it significantly restricts the period during which captured authentication assertions remain useful.

Consequently, adversaries must operate with substantially greater timing precision while legitimate users remain protected from normal network variability.

Infrastructure Reconnaissance

Many successful attacks begin with prolonged observation of infrastructure behaviour before active exploitation occurs.

Static service locations, predictable deployment patterns, and long-lived network topologies may allow adversaries to perform detailed reconnaissance over extended periods before launching an attack.

The adaptive infrastructure concepts explored within the PaCoDu framework seek to reduce this predictability through infrastructure mutation, moving-target defence concepts, and deception-oriented deployment strategies.

These mechanisms do not prevent reconnaissance altogether. Instead, they aim to reduce its efficiency by increasing uncertainty and requiring repeated analysis as infrastructure characteristics evolve.

Deception-Layer Analysis

Defensive deception introduces its own challenges.

Given sufficient observation time and data collection capability, sophisticated adversaries may attempt statistical analysis to distinguish genuine infrastructure components from intentionally deceptive resources.

Large-scale correlation analysis, behavioural fingerprinting, or long-term infrastructure mapping may eventually reduce the effectiveness of deception if defensive strategies remain insufficiently dynamic.

Accordingly, deception should not be viewed as a permanent protective mechanism but as one component within a broader layered defensive architecture.

Its principal value lies in increasing adversarial uncertainty, consuming attacker resources, and improving defender visibility during reconnaissance and intrusion attempts.

Post-Authentication Compromise

Strong authentication cannot eliminate risks that occur after identity verification has been successfully completed.

Endpoint malware, session hijacking, browser compromise, credential forwarding, memory scraping, and privilege escalation remain possible even when authentication itself functions correctly.

The PaCoDu framework therefore recognises authentication as only one component of overall system security.

Secure session management, endpoint protection, continuous monitoring, behavioural anomaly detection, and least-privilege access control remain necessary complementary defensive mechanisms.

Artificial Intelligence as an Adversarial Force

Artificial intelligence is expected to influence both defensive and offensive cybersecurity capabilities.

Automated reconnaissance, behavioural modelling, phishing generation, vulnerability discovery, and attack orchestration may increasingly reduce the time required for adversaries to identify potential weaknesses.

At the same time, defensive systems may employ AI-assisted monitoring, anomaly detection, behavioural analysis, and automated incident response to improve defensive effectiveness.

The PaCoDu framework remains intentionally neutral regarding this technological evolution.

Rather than relying upon artificial intelligence as a primary security mechanism, the framework is designed so that its architectural principles remain applicable regardless of whether future attacks are conducted primarily by human operators, automated systems, or collaborative human-AI teams.

Security Perspective

The PaCoDu Vault should not be interpreted as claiming resistance against every attack class described above.

Instead, the framework proposes that defensive architectures should be evaluated according to the cumulative resources required for successful compromise.

Even when individual defensive layers can eventually be defeated, increasing the total investment required across multiple independent attack dimensions may substantially reduce the practical attractiveness of attacking the system.

Within this perspective, resilience is understood not as the absence of compromise, but as the progressive escalation of attacker cost throughout the complete authentication lifecycle.

8. OPERATIONAL LIMITATIONS AND RESEARCH BOUNDARIES

The PaCoDu Vault is presented as an AI-assisted conceptual architectural framework rather than a finished authentication protocol or production-ready security standard.

Although the framework integrates established security concepts drawn from modern authentication research, it does not claim that their combination has been formally validated through large-scale implementation, cryptographic proof, or operational deployment.

Accordingly, the following limitations define the intended scope of this work.

No Claim of Perfect Security

The PaCoDu framework does not claim to create an authentication system that is impossible to compromise.

The underlying philosophy of this research begins with the opposite assumption: every sufficiently valuable system can eventually be attacked given enough time, expertise, computational capability, financial investment, and organisational resources.

Rather than attempting to eliminate this reality, the framework investigates whether authentication architectures can be designed to progressively increase the cumulative cost of successful compromise.

The objective is therefore deterrence through cost escalation rather than the pursuit of absolute security.

No Novel Cryptographic Primitive

The framework does not introduce a new encryption algorithm, hashing function, digital signature scheme, or cryptographic primitive.

Instead, PaCoDu combines established security mechanisms—including HMAC-based authentication, fuzzy extractors, hardware-protected execution environments, behavioural authentication, moving-target defence concepts, and deception-oriented infrastructure—within a unified architectural model.

The conceptual contribution lies in the integration and coordination of these mechanisms toward a common defensive objective rather than in the invention of new mathematical cryptography.

Implementation Independence

The architecture intentionally avoids prescribing specific implementation technologies.

The authentication pipeline may be implemented using different hardware platforms, operating systems, communication protocols, biometric modalities, or infrastructure designs depending upon operational requirements.

Similarly, temporal authentication intervals, behavioural authentication methods, infrastructure mutation strategies, and deception mechanisms should be regarded as engineering parameters rather than mandatory protocol specifications.

This flexibility allows the conceptual architecture to remain applicable across diverse deployment environments while encouraging future implementation research.

Research Rather Than Product Specification

The PaCoDu Vault should not be interpreted as a commercial product specification, implementation blueprint, or security certification framework.

Several architectural components remain intentionally conceptual and require further investigation through prototype development, experimental validation, usability studies, formal cryptographic analysis, and independent peer review before practical deployment could be considered.

Future work should evaluate both the technical feasibility and operational effectiveness of the proposed architecture under realistic deployment conditions.

AI-Assisted Research Methodology

Artificial intelligence played a significant role throughout the development of this work.

AI systems assisted in refining terminology, improving technical clarity, exploring alternative architectural structures, reviewing logical consistency, strengthening mathematical presentation, and identifying weaknesses requiring clarification.

The central architectural philosophy, systems-level reasoning, and overall conceptual direction originated from the author, while AI served as an advanced collaborative research and editorial tool.

Accordingly, the PaCoDu Vault should be understood as an example of AI-assisted conceptual systems research rather than AI-generated autonomous design.

Intended Contribution

The contribution of this paper should not be evaluated according to whether every proposed mechanism is individually novel.

Instead, the intended contribution is the presentation of a coherent architectural philosophy that reframes authentication security as a problem of economic deterrence.

Rather than asking, "Can a system be made impossible to hack?", the framework asks a different question:

"Can authentication architectures be designed so that successful compromise becomes progressively less attractive because the required investment in time, expertise, equipment, organisational coordination, and operational risk exceeds the expected benefit?"

Whether this hypothesis ultimately proves correct remains an open question requiring future academic investigation.

The purpose of this work is to establish that question as a legitimate direction for further research.

9. CONCLUSION — TOWARD A COST-ESCALATION PHILOSOPHY OF AUTHENTICATION

The PaCoDu Vault originated from a simple observation.

Modern authentication systems have achieved remarkable levels of technical sophistication, yet they continue to operate within a common philosophical objective: preventing unauthorized access by making systems increasingly difficult to compromise.

This objective has produced significant advances in cryptography, identity management, hardware security, and distributed authentication. However, the complete elimination of compromise remains an unrealistic expectation for sufficiently valuable systems.

The PaCoDu framework therefore begins from a different assumption.

Rather than asking whether an authentication system can be made impossible to hack, it asks whether successful compromise can be made progressively less attractive by continuously increasing the cumulative resources required to achieve it.

Within this perspective, security is no longer viewed exclusively as a binary state in which a system is either secure or compromised. Instead, security becomes an economic and operational process in which every defensive layer contributes to increasing the cost of attack.

The framework proposes that authentication architectures should be evaluated not only according to their cryptographic strength, but also according to the cumulative investment required of a successful adversary.

These investments include, but are not limited to:

- Time required for reconnaissance and exploitation.
- Financial investment in specialised hardware and infrastructure.
- Computational resources necessary to sustain prolonged attacks.
- Specialist expertise spanning multiple technical disciplines.
- Organisational coordination between multiple participants.
- Operational exposure and probability of detection.

Under this philosophy, compromise may remain theoretically possible, but its practical attractiveness decreases as each additional defensive layer consumes greater resources while providing progressively smaller returns.

Consequently, the objective of the defender shifts from pursuing unattainable perfection toward influencing attacker decision-making.

A rational adversary continuously evaluates the expected value of an attack against the anticipated investment required to complete it. If the required expenditure of time, equipment, expertise, organisational effort, and operational risk consistently exceeds the expected reward, many attacks become economically irrational regardless of whether they remain technically feasible.

This perspective represents the central conceptual contribution of the PaCoDu Vault.

The framework does not claim to introduce new cryptographic mathematics, new authentication primitives, or an unbreakable security protocol.

Instead, it proposes an alternative architectural philosophy in which established defensive mechanisms are intentionally combined to increase cumulative attacker cost across multiple independent dimensions.

Whether this philosophy ultimately proves effective remains a question for future research, experimental validation, prototype implementation, and independent academic review.

Nevertheless, if the ideas presented within this paper encourage researchers to evaluate authentication systems not only according to resistance against compromise but also according to their ability to influence attacker economics, operational behaviour, and strategic decision-making, then the primary objective of this work will have been achieved.

The PaCoDu Vault is therefore presented not as the conclusion of a research programme, but as the beginning of a broader discussion regarding how future authentication architectures might evolve beyond the pursuit of absolute security toward the deliberate engineering of economic deterrence.

PaCoDu Design Principle

Perfect security is an unattainable objective.

Practical security is achieved when the cumulative cost of successful compromise exceeds the expected value of the attack.

The purpose of authentication is therefore not to eliminate attackers, but to influence their decisions by transforming compromise into an economically irrational undertaking.

ACKNOWLEDGEMENTS

The author wishes to acknowledge the significant role that artificial intelligence played throughout the development of this work.

The PaCoDu Vault originated as an independent systems-level concept developed through first-principles reasoning regarding authentication security. During its evolution, AI-assisted collaboration contributed substantially to the refinement of terminology, architectural organisation, technical writing, mathematical presentation, structural consistency, critical analysis, and the identification of conceptual limitations.

Artificial intelligence did not replace the author's role in originating the central architectural philosophy. Instead, it functioned as an advanced collaborative research assistant capable of accelerating interdisciplinary exploration and improving technical communication.

The author also recognises the broader cybersecurity and cryptographic research community whose published work established many of the foundational concepts upon which this conceptual framework builds.

Any remaining errors, omissions, or weaknesses remain the sole responsibility of the author.

FUTURE RESEARCH DIRECTIONS

The PaCoDu Vault intentionally presents a conceptual architectural framework rather than a completed engineering solution. Consequently, numerous opportunities exist for future investigation.

Potential research directions include:

- Development of a working proof-of-concept implementation.
- Formal security analysis using accepted cryptographic methodologies.
- Usability evaluation of multimodal behavioural authentication.
- Performance benchmarking across geographically distributed infrastructures.
- Quantitative modelling of attacker cost using economic and game-theoretic approaches.
- Investigation of adaptive deception strategies under prolonged adversarial observation.

- Evaluation of AI-assisted defensive orchestration using existing cybersecurity knowledge bases and automated response systems.
- Comparative analysis against conventional authentication architectures using measurable security metrics.
- Investigation of optimal temporal authentication intervals under realistic deployment conditions.
- Independent peer review and academic evaluation of the proposed cost-escalation philosophy.

I hope that future researchers may choose either to validate, challenge, refine, or reject elements of the framework through rigorous experimental investigation.

REFERENCES

The PaCoDu Vault builds upon numerous established concepts from authentication, cryptography, cybersecurity, distributed systems, and hardware security research.

Representative foundational references include:

NIST Special Publication 800-63B — Digital Identity Guidelines: Authentication and Lifecycle Management.

NIST Special Publication 800-207 — Zero Trust Architecture.

Ross Anderson. *Security Engineering: A Guide to Building Dependable Distributed Systems*.

Bruce Schneier. *Applied Cryptography*.

Menezes, van Oorschot and Vanstone. *Handbook of Applied Cryptography*.

Trusted Computing Group (TCG) Specifications.

Intel Software Guard Extensions (Intel SGX) Documentation.

ARM TrustZone Documentation.

Yevgeniy Dodis et al. "Fuzzy Extractors: How to Generate Strong Keys from Biometrics and Other Noisy Data."

Research literature relating to Moving Target Defense (MTD).

Research literature relating to Honeytokens and Deception Technology.

Research literature relating to Behavioural Biometrics.

Research literature relating to Hardware Security Modules (HSMs).

Research literature relating to Trusted Execution Environments (TEEs).

Research literature relating to Human-Centred Authentication Systems.

Artificial intelligence systems were used throughout the preparation of this manuscript to improve technical writing, editorial consistency, architectural analysis, and conceptual

development. AI-assisted contributions are described explicitly within the paper as part of the documented research methodology.

FINAL AUTHOR'S STATEMENT

The PaCoDu Vault should not be interpreted as a claim of having solved authentication.

Instead, it represents the work of an independent systems thinker who questioned a long-standing assumption.

If the ideas presented here stimulate discussion, invite criticism, encourage experimentation, or inspire future researchers to investigate authentication through the lens of attacker economics rather than absolute prevention, then this project will have fulfilled its intended purpose.

Every mature scientific idea begins as a question before it becomes an answer.

The PaCoDu Vault is offered in that spirit.

APPENDIX A — AI-ASSISTED RESEARCH METHODOLOGY

A.1 Purpose

The PaCoDu Vault originated as an independent conceptual investigation into authentication security and was subsequently developed through an iterative collaboration between the author and advanced artificial intelligence systems.

This appendix is provided in the interest of methodological transparency. Its purpose is to distinguish the author's original intellectual contribution from the role performed by AI throughout the research and writing process.

A.2 Origin of the Concept

The central research question originated independently from the author:

"If authentication systems cannot realistically be made impossible to compromise, can they instead be designed so that successful compromise becomes progressively more expensive, resource-intensive, and operationally unattractive?"

This question ultimately evolved into the cost-escalation philosophy described throughout the PaCoDu framework.

The architectural direction, systems-level reasoning, and conceptual objectives originated from the author and were refined through iterative analysis rather than generated autonomously by AI.

A.3 Role of Artificial Intelligence

Artificial intelligence served as an advanced collaborative research assistant throughout the development of this work.

Its primary contributions included:

- Improving technical writing and academic style.
- Refining architectural organisation.
- Increasing logical consistency between chapters.

- Suggesting clearer terminology.
- Reviewing mathematical notation.
- Identifying conceptual weaknesses and unsupported claims.
- Challenging assumptions and encouraging clarification.
- Exploring alternative architectural interpretations.
- Assisting with literature awareness and technical context.

AI was not used as a substitute for independent reasoning or original conceptual development.

Instead, it functioned as an interactive analytical tool that accelerated interdisciplinary exploration while leaving all final design decisions to the author.

A.4 Human Responsibility

Although AI significantly assisted the development process, responsibility for the content of this paper remains entirely with the author.

This includes:

- The selection of architectural components.
- The overall research direction.
- The interpretation of existing security concepts.
- The formulation of the cost-escalation philosophy.
- The final acceptance or rejection of AI-generated suggestions.

Any remaining errors, omissions, unsupported assumptions, or technical inaccuracies remain solely the responsibility of the author.

A.5 Methodological Perspective

The PaCoDu Vault should therefore be understood as an example of AI-assisted conceptual systems research.

Rather than replacing human creativity, AI functioned as an intellectual amplifier capable of accelerating idea refinement, improving technical communication, and supporting interdisciplinary synthesis.

The author believes that transparent human-AI collaboration represents a legitimate research methodology when the origin of ideas, the role of AI, and the limits of the resulting work are clearly documented.

Appendix A Conclusion

The development of the PaCoDu Vault demonstrates that artificial intelligence can function as a collaborative research partner without replacing human authorship.

The framework presented in this paper reflects the author's original conceptual direction, strengthened through iterative AI-assisted analysis, technical refinement, and critical review.

Accordingly, the PaCoDu Vault should be interpreted not as an AI-generated proposal, but as an AI-assisted conceptual architectural framework whose intellectual responsibility remains entirely with the author.

APPENDIX B — AI-ASSISTED ADAPTIVE DEFENSIVE ORCHESTRATION

B.1 Purpose

The PaCoDu Vault proposes an authentication architecture whose primary objective is to increase the cumulative cost of successful compromise. While the core framework focuses on identity verification and adaptive defensive layers, future implementations may benefit from intelligent orchestration systems capable of responding to security events at machine speed.

This appendix explores one possible extension of the framework through the integration of an AI-assisted defensive orchestration component.

The concepts presented here remain speculative and are intended as future research directions rather than implementation requirements.

B.2 Design Philosophy

The proposed AI component is not intended to invent new defensive mechanisms or autonomously redesign system security.

Instead, its role is comparable to that of an experienced security analyst capable of rapidly selecting appropriate responses from an existing body of defensive knowledge.

Conceptually, the system performs four sequential functions:

1. Detect abnormal behaviour or suspected malicious activity.
2. Classify the observed attack according to known threat categories.
3. Select the most appropriate defensive response from an established knowledge base.
4. Coordinate the execution of existing security controls.

The intelligence therefore resides primarily in the speed and consistency of defensive decision-making rather than in autonomous invention.

B.3 Operational Workflow

A conceptual operational sequence may be represented as follows:

Normal Activity



Continuous Telemetry Collection



Threat Detection



Attack Classification



Knowledge Base Query



Selection of Defensive Response



Execution of Existing Security Controls



Continuous Monitoring and Reassessment

Each stage relies upon information already available to the defensive infrastructure rather than requiring speculative predictive capabilities.

B.4 Knowledge-Based Defensive Selection

Rather than generating novel countermeasures, the AI component maintains a structured repository of documented attack categories together with corresponding defensive procedures.

Examples may include:

- Credential stuffing.
- Password spraying.
- Replay attempts.
- Distributed denial-of-service attacks.
- Session hijacking.
- Automated reconnaissance.
- Brute-force authentication attempts.
- Bot-driven abuse.
- Behavioural anomalies.
- Infrastructure scanning.

When a threat is detected, the orchestration engine compares observed characteristics against this knowledge base and selects defensive actions according to predefined security policies.

Possible responses may include:

- Dynamic rate limiting.
- Temporary account protection.
- Additional authentication requirements.
- Infrastructure rerouting.
- Deployment of deception resources.
- Network isolation.
- Alert generation.
- Human analyst notification.

The AI therefore acts as an intelligent coordinator rather than an autonomous security decision-maker.

B.5 Relationship to the PaCoDu Philosophy

Within the PaCoDu framework, AI-assisted orchestration contributes to cost escalation in two important ways.

First, rapid defensive response reduces the amount of time available for attackers to exploit newly discovered opportunities before defensive controls are activated.

Second, adaptive response introduces uncertainty into attacker planning because defensive actions may vary according to observed behaviour rather than remaining completely static.

These characteristics increase operational complexity without requiring the invention of entirely new defensive technologies.

B.6 Research Challenges

Several important research questions remain unresolved.

These include:

- False-positive detection rates.
- Knowledge base maintenance.
- AI model transparency and explainability.
- Adversarial manipulation of machine-learning systems.
- Performance under large-scale distributed attacks.
- Human oversight of automated defensive decisions.

These challenges require substantial investigation before practical deployment could be considered.

Appendix B Conclusion

The AI-assisted orchestration component presented in this appendix should be understood as an optional extension to the PaCoDu architecture rather than a fundamental requirement.

Its purpose is not to replace existing security mechanisms or human decision-making, but to improve the speed, consistency, and coordination with which established defensive responses are selected and executed.

Within the broader PaCoDu philosophy, artificial intelligence serves as a force multiplier that accelerates defensive operations while supporting the central objective of progressively increasing the cumulative cost of successful compromise.

APPENDIX C — TEMPORAL AUTHENTICATION SIMULATION

C.1 Purpose

The PaCoDu Vault is presented as a conceptual authentication architecture rather than a completed implementation. Consequently, the objective of this appendix is not to validate the framework as a whole, but to demonstrate the operational behaviour of one specific component: the temporal authentication mechanism described in Section 6.

The simulation investigates whether a short-lived authentication window remains operationally viable under representative network latency conditions and illustrates the purpose of the bounded sliding validation window introduced in Stage 2.1.

Accordingly, the results presented here should be interpreted as an engineering demonstration rather than a formal security proof.

C.2 Simulation Objectives

The simulation was designed to evaluate three operational questions:

- Can short-lived authentication tokens be generated deterministically from a shared secret and a common temporal reference?
- Can normal network latency be accommodated without causing unnecessary rejection of legitimate authentication attempts?
- Does the bounded sliding validation window improve operational robustness while preserving the short-lived nature of authentication assertions?

The simulation intentionally focuses on functional behaviour rather than cryptographic strength.

C.3 Methodology

A lightweight Python simulation was developed to reproduce the temporal authentication logic described within the main body of this paper.

The simulation performs the following sequence:

1. Generate a temporal authentication token from a shared secret using an HMAC-SHA-512 construction.
2. Simulate transmission across a network experiencing variable propagation delay.
3. Calculate the server-side temporal epoch.
4. Verify the received token against both the current temporal window (T) and the immediately preceding window ($T-1$).

The implementation uses a conceptual 1.5-second temporal interval solely to demonstrate the behaviour of the authentication pipeline. This value should not be interpreted as a universally optimal deployment parameter.

C.4 Conceptual Authentication Logic

The temporal authentication process implemented by the simulation may be represented conceptually as follows:

```
FUNCTION GenerateTemporalToken(secret_key)

    current_time ← CurrentUnixTimestamp()

    time_step ← floor(current_time / Window_Size)

    token ← HMAC_SHA_512(secret_key, time_step)

    RETURN token
```

Both the client and the verification server independently derive the same temporal value from their respective system clocks. Successful authentication therefore depends upon both parties calculating compatible temporal windows while tolerating minor network variation.

C.5 Python Reference Implementation

The following reference implementation was used to reproduce the conceptual behaviour of the temporal authentication pipeline.

```
import hmac
import hashlib
import math
import random

class PaCoDuSimulation:

    def __init__(self, key):
        self.key = key.encode()

    def get_step(self, seconds):
        return math.floor(seconds / 1.5)

    def make_token(self, step):
        return hmac.new(
            self.key,
            str(step).encode(),
            hashlib.sha512
        ).hexdigest()

    def verify_token(self, token, server_time):

        current = self.get_step(server_time)
        previous = current - 1

        if token == self.make_token(current):
            return "SUCCESS: Valid inside current window."

        if token == self.make_token(previous):
            return "SUCCESS: Valid inside bounded lookback window."

        return "REJECTED: Token expired or invalid."

simulation = PaCoDuSimulation(
    "PACODU_SECRET_SEED_STRING"
)

base_time = 1800000000.0

client_time = base_time + 1.48

client_step = simulation.get_step(client_time)

token = simulation.make_token(client_step)

network_delay = random.uniform(0.15, 0.40)

server_time = client_time + network_delay

print(
    simulation.verify_token(
        token,
        server_time
    )
)
```

C.6 Representative Test Scenario

To demonstrate the effect of network latency, a representative scenario was executed.

The client generated an authentication token approximately 20 milliseconds before the temporal window advanced.

A simulated propagation delay between 150 and 400 milliseconds was then introduced before the token reached the verification server.

Under a strict single-window validation model, the authentication attempt would frequently fail because the server had already progressed into the next temporal epoch.

By validating against both T and $T-1$, the verification process successfully tolerated normal network variability while maintaining short credential lifetimes.

C.7 Interpretation of Results

The simulation demonstrates that bounded temporal tolerance significantly improves operational reliability without requiring long-lived authentication assertions.

Importantly, this should not be interpreted as proof that the complete PaCoDu architecture has been experimentally validated.

Instead, the simulation confirms only that the proposed temporal authentication mechanism behaves consistently under representative latency conditions and that the sliding validation window performs the engineering function for which it was designed.

Further validation would require prototype implementation, distributed deployment, large-scale performance testing, usability evaluation, and independent security analysis.

Appendix C Conclusion

The simulation presented in this appendix provides an operational demonstration of one individual component of the PaCoDu authentication pipeline.

Although limited in scope, the results support the engineering rationale for combining high-frequency temporal authentication with a bounded sliding validation window capable of absorbing realistic network latency.

The appendix therefore illustrates that the proposed mechanism is operationally plausible within a controlled software environment while reinforcing the broader philosophy of the PaCoDu framework: increasing attacker cost without unnecessarily compromising usability for legitimate users.

Future research should expand this work through prototype implementations, formal verification, and large-scale experimental evaluation under realistic deployment conditions.

APPENDIX D — THE COST-ESCALATION SECURITY MODEL

D.1 Purpose

The central contribution of the PaCoDu Vault is not the invention of a new cryptographic algorithm, authentication protocol, or security primitive.

Instead, the framework proposes an alternative perspective for evaluating authentication architectures.

This appendix formalises the underlying design philosophy introduced throughout the paper by describing security as a process of increasing the cumulative cost required for successful compromise rather than pursuing the unattainable objective of absolute protection.

D.2 Traditional Security Perspective

Historically, authentication systems have been designed around a common objective:

Prevent unauthorised access by making systems increasingly difficult to compromise.

This philosophy has produced significant advances in cryptography, hardware security, identity management, access control, and distributed authentication.

Despite these advances, no practical security system can reasonably be expected to remain permanently resistant against sufficiently capable adversaries with unlimited resources and sufficient time.

Consequently, the PaCoDu framework begins from a different premise.

Rather than attempting to eliminate compromise entirely, it seeks to influence attacker decision-making by continuously increasing the resources required for successful compromise.

D.3 The Cost-Escalation Principle

Within the PaCoDu framework, security is viewed as the cumulative effect of multiple defensive layers that progressively increase attacker investment across several independent dimensions.

Conceptually,

Authentication Security $\approx$ Progressive Cost Escalation

where attacker cost is influenced by the combined effect of:

Attacker Cost = f(Time, Equipment, Expertise, Computational Resources, Financial Investment, Coordination, Operational Risk)

No individual factor is assumed to prevent compromise independently.

Instead, resilience emerges from the combined increase across all contributing dimensions.

D.4 Dimensions of Attacker Cost

The PaCoDu framework identifies several representative dimensions that may influence the practical attractiveness of an attack.

Time

The duration required to complete reconnaissance, preparation, exploitation, persistence, and successful compromise.

Financial Investment

The monetary resources required to obtain specialised hardware, infrastructure, software, laboratory facilities, or operational support.

Computational Resources

The processing capability required to sustain prolonged cryptographic analysis, distributed attacks, behavioural modelling, or automated reconnaissance.

Technical Expertise

The specialist knowledge necessary to understand, analyse, and defeat multiple independent security mechanisms.

Organisational Coordination

The number of individuals, disciplines, or coordinated activities required to execute successful compromise.

Systems requiring collaboration between multiple specialists generally present higher operational complexity than those susceptible to individual attackers.

Operational Risk

The probability that attacker activity will be detected, interrupted, attributed, or otherwise exposed before successful compromise is achieved.

D.5 Defender Objective

Under the PaCoDu philosophy, the objective of defensive design is not necessarily to maximise any single security mechanism.

Instead, the objective is to maximise the cumulative resources required for successful compromise while preserving acceptable usability for legitimate users.

Conceptually,

Defender Success $\propto$ Attacker Cost / Expected Reward

As attacker cost increases while expected reward remains constant, the practical attractiveness of compromise decreases.

Although technically possible attacks may continue to exist, many become economically irrational.

D.6 Relationship to the Five-Ring Architecture

Each defensive ring described within the main body of this paper contributes to one or more dimensions of attacker cost.

Ring 1 primarily increases reconnaissance effort and infrastructure requirements.

Ring 2 increases behavioural analysis complexity and observation time.

Ring 3 increases hardware requirements, equipment sophistication, and technical expertise.

Ring 4 reduces credential lifetime while increasing timing precision and replay complexity.

Ring 5 increases uncertainty, reconnaissance effort, operational workload, and probability of detection.

The architecture therefore functions as a coordinated system whose cumulative objective is the progressive escalation of attacker investment.

D.7 Relationship to Existing Security Models

The PaCoDu framework should not be interpreted as replacing established cybersecurity principles such as defence-in-depth, Zero Trust, adaptive authentication, moving-target defence, or hardware-rooted trust.

Instead, it provides a complementary conceptual perspective through which these established mechanisms may be viewed as contributors to a broader economic deterrence strategy.

Future research may investigate whether attacker cost can be quantified more formally using methods drawn from economics, decision theory, operational research, or game theory.

Such work lies beyond the scope of the present paper but represents a promising direction for continued investigation.

D.8 Closing Perspective

The PaCoDu Vault does not propose that security can be made absolute.

Instead, it proposes that authentication systems should increasingly be evaluated according to the cumulative resources required for successful compromise.

When successful attacks demand prolonged effort, specialised expertise, expensive equipment, coordinated teams, significant computational capability, and increased operational risk, many opportunities that remain technically feasible may cease to be practically attractive.

In this sense, the objective of authentication evolves from preventing every attack to influencing attacker behaviour through deliberate cost escalation.

Appendix D Conclusion

The Cost-Escalation Security Model provides the theoretical foundation upon which the PaCoDu Vault is built.

Rather than redefining cryptography itself, the framework proposes a different way of evaluating authentication architectures—one that considers not only resistance to compromise, but also the economic, operational, and organisational burden imposed upon potential adversaries.

Whether this philosophy proves broadly applicable remains a question for future academic investigation. Nevertheless, it offers a conceptual lens through which authentication may be viewed not solely as a technical discipline, but also as an exercise in influencing adversarial decision-making through the deliberate engineering of cumulative cost.